

(SCHOOL OF ENGINEERING)
Mangaluru
SCHOOL OF ENGINEERING
(UG Programme)
II Internal Exam - November 2025
BTech AIML/ECE - Semester I
Linear Algebra and Calculus

Time: 1½ hrs.

Max Marks: 50

Note:

1. This question paper consists of 2 sections. PART- A and PART- B.
2. Answer FOUR questions from PART B either 8 OR 9 and 10 OR 11.

PART - A

I Answer any **FIVE** of following questions:

(5 x 2 =10)

- Illustrate the Cartesian equation from the parametric form of the curve $x =$
1. $3t, y = 9t^2, -\infty < t < \infty$. $y = (3t)^2 = x^2 \therefore y = x^2 \rightarrow \text{Parabola}$
 2. Solve for the angle of intersection of the curves $r = \sin\theta + \cos\theta, r = \sin\theta$. $\psi_1 \neq \psi_2$
 3. Describe the radius of curvature of a straight line. $\rho = \frac{1}{0} \rightarrow \text{infinity}$
 4. State Clairaut's theorem and use for $f(x, y) = \tan^{-1}\frac{y}{x} - y^2 \tan^{-1}\frac{x}{y}$ $\left(\frac{1}{1+x^2}\right)$
 5. Identify the extrema of $f(x, y) = x^3 y^2 (1 - x - y)$.
 6. Evaluate $\int_0^4 \int_0^x x(x^2 + y^2) dx dy$.
 7. By changing the order of integration, evaluate $\int_0^3 \int_1^{\sqrt{4-y}} (x + y) dx dy$

PART - B

(4 x 10 =40)

II Answer the following:

- a. Evaluate the length of the asteroid $x = \cos^3 t, y = \sin^3 t, 0 \leq t \leq \frac{\pi}{2}$. (10)
8. b. Solve for the dimensions of the rectangular box, open at the top, of maximum capacity whose surface is 432 sq.cm. (10)

OR

- a. Evaluate the arclength of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ in the first quadrant. (10)
9. b. If $v = (x^2 + y^2 + z^2)^{-\frac{1}{2}}$, then discuss about the value of $v_{xx} + v_{yy} + v_{zz}$. (10)

III Answer the following:

- a. Evaluate $\iint_A xy dx dy$ where A is the domain bounded by x -axis, ordinate $x = 6$ and the curve $x^2 = 12y$. Sketch the domain A . (10)
10. b. Evaluate $\iint r \sin \theta dr d\theta$ over the cardioid $r = 2(1 - \cos \theta)$ above the initial line. Sketch the curve and the area mentioned. (10)

OR

11. Evaluate $\iint_R x^3 dx dy$ where R is the region in the first quadrant
 a. boundd by the lines $x = y, y = 0, x = 8$ and the curve $xy = 16$. (10)
 Sketch the region R .
 b. Evaluate $\iint r^3 dr d\theta$ over the area included between the circles $r = \sin\theta$ and $r = 2\sin\theta$. Sketch the curve and the area mentioned. (10)

Date: 03.11.2025

Time: 10.00-11.30

Copies: 110

Mapping of Questions with Bloom's Level and Course Outcomes (COs)

Part	A							B							
Question	I							II							
Q. No.	1	2	3	4	5	6	7	8a	8b	9a	9b	10a	10b	11a	11b
Blooms Level	4	3	3	3	3	5	5	3	3	3	3	3	3	3	3
Mapped CO(s)	3	3	3	3	3	4	4	3	3	3	3	4	4	4	4